

Dampers:

Dampers are crucial components in HVAC systems. They are essentially valves that control the flow of air within the system, offering numerous functionalities like:

- **Directing airflow:** Damper positioning helps direct air to specific areas or rooms in a building.
- **Regulating flow rates:** By adjusting the damper opening, you can control the volume of air passing through a duct or plenum.
- **Maintaining pressure:** Dampers can help maintain desired pressure levels within different parts of the HVAC system.
- **Preventing backdraft:** They prevent unwanted airflow moving in the opposite direction, improving system efficiency and safety.
- **Isolating areas:** In case of fire or smoke, specific dampers can close to isolate sections of the system, preventing contamination and spread.



Main Types:

- **Single-Direction Dampers:** Functioning like non-return valves, these dampers permit air movement in a designated direction while blocking it in the opposite direction. They're typically installed downstream of fans to prevent backdraft, safeguarding fans from potential damage.
- **Pressure Relief Damper:** Maintains safe pressure levels by opening when pressure exceeds a set limit.
- **Volume Control Dampers:** Regulates airflow within the system, used to adjust the amount of air flowing through individual duct branches, controlling the temperature in specific rooms. Available with manual or motorized control options.
- **Fire and Smoke Dampers:** barriers close off ducts and isolate specific sections of a building in the event of a fire or smoke, helping contain the threat and protect people from harm.



BDD



VCD



FDI

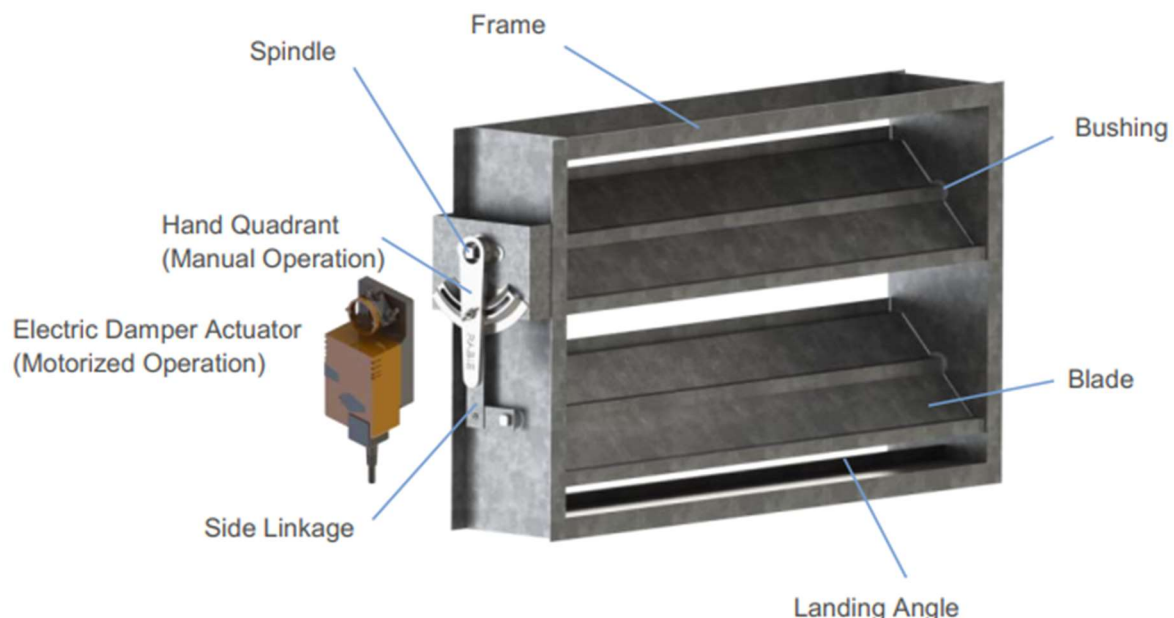
Common damper materials:

- **Galvanized steel:** Most common, affordable, offers good durability and corrosion resistance.
- **Aluminum:** Lighter weight, good for low-pressure applications, more susceptible to corrosion.
- **Stainless steel:** Highest durability and corrosion resistance, suitable for harsh environments or food processing.

Construction Details and Main Components:

While damper designs vary depending on their function and application, most share some common components:

- **Frame:** Damper backbone, provides structural support, typically made of sheet metal formed into channels or rectangular shapes.
(**Frame designs:** Wall-mounted, duct-mounted, and sleeve-mounted, Flanged, With duct mate flange, C clip, with round or oval attachment), different options suitable for various installation requirements.)
- **Blades:** Control airflow by opening and closing, made of the same material as the frame. Maybe single blade or multi blade construction or maybe a curtain blade regarding the application and size of the damper.
(**Blade types:** Airfoil, butterfly, 2-V, 3-V or Curtain. Blade variations offer different airflow characteristics and pressure drops.)
- **Blade Action:** Opening act, Some blades open and close in the same direction -Parallel blades-(standard) or in opposite directions (opposed blades).
- **Pivot points:** Location where the blades are fixed to the frame, just like hinges t these pivot points allow the blades to rotate within the frame, influencing the direction and volume of air passing through. Different placement points locations (edge, center, off-center).
- **Linkage:** physical connection between blades, allowing them to move together, Options: face linkage or side linkage.
- **Bearings:** Blade mountings, reduce friction and allow smooth blade movement, crucial for optimal operation (Nylon (for Std. control applications) or Bronze (For fire application) or brass for round dampers).
- **Control system:** Hand quadrant for manual control or Actuator for automatic control.
- **Fusible link (optional for Fire application):** Triggers automatic closure in fire dampers, melts at high temperatures.
- **Seals (optional for low-leakage applications):** Maintain airtight closure, often used in fire and smoke dampers, made of rubber or silicone.



Single-Direction Dampers

These dampers act as non-return valves, used to convey air in a specific direction and prevent it from flowing in the opposite direction. Functioning like non-return valves, typically installed downstream of fans to prevent backdraft, safeguarding fans from potential damage. They are divided into three types according to desired pressure application (Light-Medium-High).

Types:

1. Gravity-Closing Dampers (Gravity Shutter (GSH))

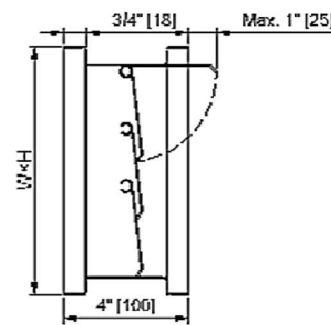
- **Application:** lightweight applications for velocities up to 1500 fpm (7.5 m/s) and pressure up to 2 In.wg (500 Pa), mainly for constant airflow systems.
- **Material:** Typically made of 1.1 mm thick extruded aluminum blades edge pivoted.
- **Frame:** There are two types of frames used for this product depending on the required installation method, as follows:

1. Wall-Mounted Frame (GSH-W): It is L-shaped as in the case of grilles to be installed on the wall opening and is also made of Extruded aluminum.

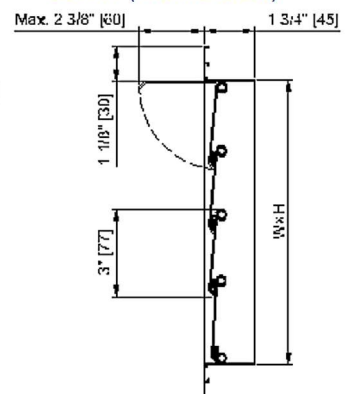
2. Duct-Mounted Frame (GSH-D): Sleeve (flat frame without a lip on the sides), made of galvanized iron for installation inside the duct.

- **Note:**
 - In the case of the wall-mounted type (GSH-W), the opening direction is to the outside of the wall unless otherwise requested.
 - This type is recommended for use with air speeds not exceeding 7.5 meters per second and static pressure not exceeding 500 Pascals.

GSH-D (Duct Mounted)



GSH-W (Wall Mounted)



2. Backdraft Dampers (for Heavy Loads) (BDD)

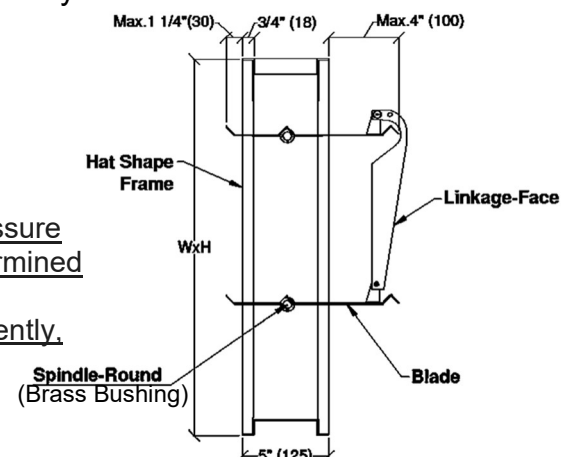
- **Application:** rated for velocities up to 2500 fpm (12.5 m/s) and pressure up to 3 in.wg (750 Pa). BDD is mainly used for variable airflow systems.
- **Material:** Typically constructed of galvanized blades, 1mm thick, with off-centered pivot at 1:3 Blade depth.
- **Frame:** The frame takes the shape of a hat to be installed inside the duct and is 1.5 mm thick for duct mounting and optionally with one side flange for wall mounting.
- **Blades:** The blades are connected together by a linkage so that they work as a single group, unlike the previous type. Each blade rests on nylon supports. They are 1 mm thick.

Note:

-The thickness of the blade can be increased to increase the pressure required for opening, and the width of the blade can also be determined by the customer.

-This product can be ordered so that each blade works independently, i.e. without linkages.

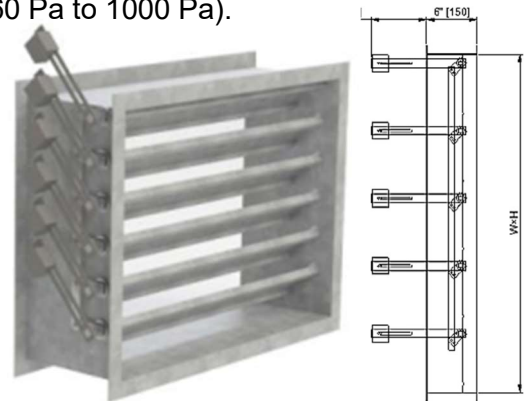
Side Linkage



3. Pressure-Operated Dampers (RDP)

Pressure relief dampers act like automatic safety valves for HVAC systems. They open at a pre-set air pressure, preventing damage to ducts and equipment. Commonly used in staircase, they maintain negative pressure for easy escape door opening during emergencies. Can be adjusted by using counter weights ranging from 0.25" w.g. to 4.0" w.g. (60 Pa to 1000 Pa).

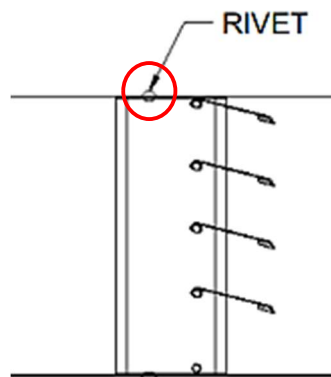
- **Application:** rated for velocities up to 3000 fpm (15.24 m/s) and pressure up to 13.5 In Wg (3.5 kPa).
- **Blades:** Airfoil shape or 2-V shape 1mm made of galvanized steel edge pivoted.
- **Frame:** galvanized steel channel, 1.5mm thick.



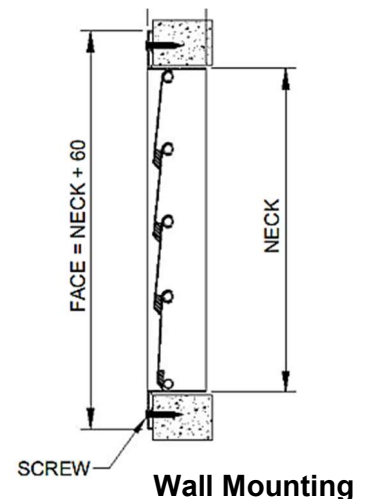
Options:

- Any modification can be made to the shape of the frame and blades made of galvanized steel and their thickness according to the customer's request.
- A bird or insect-proof mesh can be added upon request.
- The edges of the blades are equipped with a rubber strip to prevent air leakage in the closed position, and it can be made of rubber upon request.

Installation:



Duct Mounting



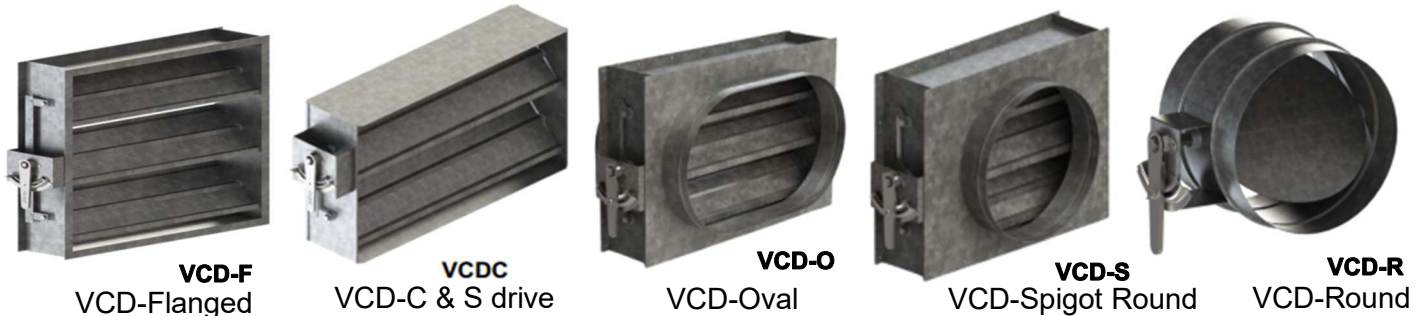
Wall Mounting

Volume Control Dampers

Volume control dampers are used to adjust the amount of air flowing through a specific duct branch in an HVAC system. They can be manually operated or motorized.

Components:

- **Frame:** The frame provides structural support for the damper and houses the blades. It is typically made of galvanized steel of 1.5mm thickness. With different frame options suitable for various installation requirements.



- **Blades:** The blades are 3-V, typically made of 1.5mm thick galvanized steel. The Damper can be provided either with single blade or with multiple blades depending on the damper height.

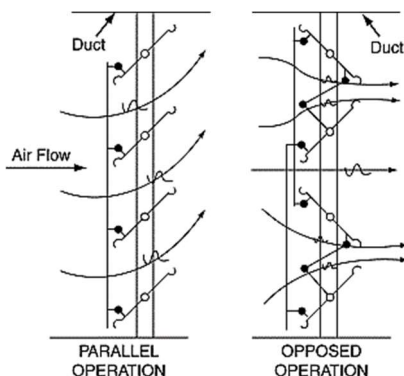
Damper Height (mm)	Number of Blades
250	1
500	2
700	3
1000	4
1200	5
1400	6
1700	7
1830	8



Single or Multi Blade Construction

We also offer 2 blades arrangements options:

- Parallel Blades (Standard)
- Opposed Blades



- Volume Damper can be Provided with any desired material or thickness based on customer's request.
- Blade Configuration and number of blades can be customized based on the customer's needs.
- Maximum Damper size as single section is (1200x1830 or 2440x1525) mm for VCD-F & 1220x1200 mm for VCD-C.
- Volume Damper can be supplied with duct mate flange for connection upon request.

Types:

There are two main types of volume control dampers:

- **Manual dampers:** These dampers are operated by hand using a lever or knob.
- **Motorized dampers:** These dampers are operated by a motor that is controlled by a signal from the HVAC system.

Manual Dampers (VCDH)

Manual dampers are the simplest type of volume control damper. They are typically used in smaller HVAC systems or in applications where automatic control is not required.

The blades of a manual damper are connected to a lever or knob that is mounted on the frame. To open the damper, the lever or knob is turned clockwise. To close the damper, the lever or knob is turned counterclockwise.



Hand Quadrant For Manual Operation

Motorized Dampers (MVCD)

Motorized dampers are more complex than manual dampers, but they offer a number of advantages, including:

- **Automatic control:** Motorized dampers can be controlled by a signal from the HVAC system, which allows them to be opened and closed automatically in response to changing conditions. We offer 3 control options:
- **Greater precision:** Motorized dampers can be opened and closed more precisely than manual dampers, which can improve the efficiency of the HVAC system.
- **Less labor:** Motorized dampers require less labor to operate than manual dampers.



Electrical Actuator For Motorized Operation

The actuator of a motorized damper can be either a direct-drive actuator or a linkage actuator.

- **Direct-drive actuators:** Direct-drive actuators are connected directly to the blades of the damper. They are typically used in dampers with a small number of blades.
- **Linkage actuators:** Linkage actuators use a linkage to connect the actuator to the blades of the damper. They are typically used in dampers with a large number of blades.

Motorized dampers can be controlled in a variety of ways, including:

- **On/off control (2-P):** On/off control is the simplest type of motorized damper control. In on/off control, the damper is either fully open or fully closed. ---Standard.
- **Three-position control (3-P):** Three-position control allows the damper to be opened, closed, or in a third position between open and closed.
- **Modulating control:** Modulating control allows the damper to be opened to any desired position.

Feedback switch: A feedback switch is a device that can be connected to a motorized damper to provide feedback to the HVAC system about the position of the damper. This feedback can be used to improve the control of the HVAC system.

- Please note that while our actuators are suitable for indoor environments, outdoor installation necessitates a protective shield. To obtain an accurate quotation for outdoor applications, please specify the requirement for an actuator shield in your request, as it will involve an additional cost.

Specifications

The specifications of a volume control damper can vary depending on the application. Some of the important specifications to consider include:

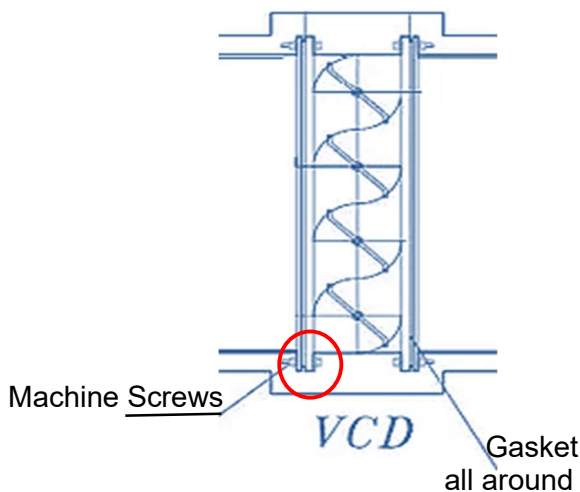
- **Size:** The size of the damper is determined by the size of the duct in which it will be installed.
- **Flow rate:** The flow rate of the damper is the maximum amount of air that it can pass.
- **Pressure drop:** The pressure drop across the damper is the amount of pressure that is lost as air flows through the damper.
- **Construction:** The construction of the damper determines its durability and performance.

Installation

The installation process will varies depending on the type of damper and the application.

VCD-F

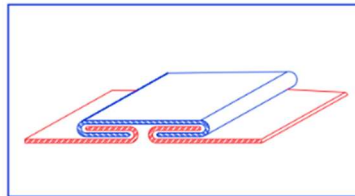
VCD with Flanged Connection



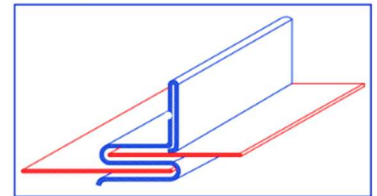
VCDC

VCD C Clip / S Slip drive connection

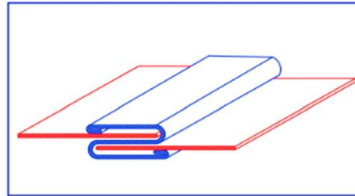
Drive Cleat (SMACNA -T1)



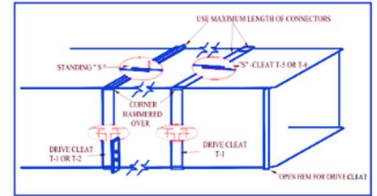
Standing "S" Cleat (SMACNA-T11)



Hemmed "S" Cleat (SMACNA T6)



Assembly:



Round & Oval Connection

Suitable with VCDR, VCDS, VCDO



Duct Mate Flange

VCD with duct mate Flange connection



You Can Scan the QR Code for more illustration.



C & S Drive Connection



Duct Mate Flange

Fire Dampers

Fire and smoke dampers are used to isolate fire or smoke or both within buildings in the event of a fire or smoke event within a specific area. Our Fire Dampers are UL certified which ensures their reliability to our customers.

Types:

- **Fire dampers:** These dampers close automatically in the event of a fire to isolate the fire and prevent it from spreading. There are mainly 2 Basic types regarding to the Blade Construction.
 - Curtain Fire Dampers.
 - Fire Damper & Blade Type.
- **Combination fire and smoke dampers:** These dampers close automatically in the event of a fire or smoke event to isolate the fire and smoke and prevent them from spreading.

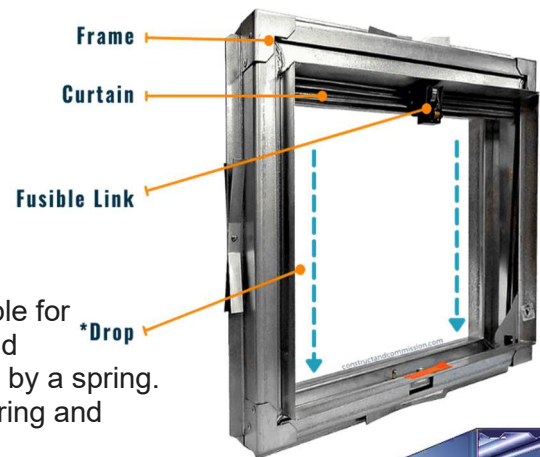


Fire dampers

There are several types of fire dampers only, each with its own advantages and disadvantages.

First: Curtain dampers

Curtain dampers are the simplest type of fire damper suitable for vertical or horizontal. They are made of galvanized steel and consist of a series of overlapping blades that are held open by a spring. In the event of a fire, the fusible link melts, releasing the spring and causing the blades to close.

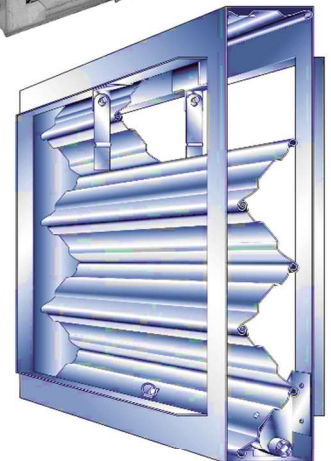


Advantages:

- Simple and inexpensive
- Easy to install
- Effective at isolating fire

Disadvantages:

- Can restrict airflow (for inside air stream type)
- Can be noisy when closing



Frame:

- Curtain dampers come with many frame options regarding the frame location with the Air stream, Various options for different free areas and to be suitable for various installation requirements.



FDI

Fire Damper Inside Air Stream



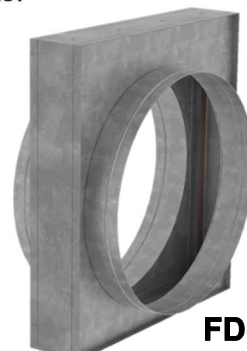
FDO

Fire Damper Outside Air Stream



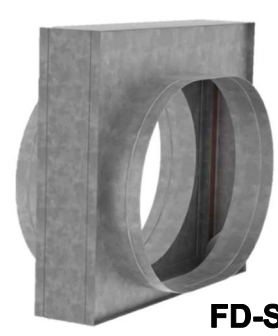
FDR

Fire Damper Rectangular



FD-SR

Fire Damper Spigot Round



FD-SO

Fire Damper Spigot Oval

Second: Fire damper Blade Type (FDB)

Blade Fire dampers are more complex than curtain dampers, but they offer a number of advantages, including:

- Less restriction of airflow
- Less noise when closing
- More durable

Blade Fire dampers consist of a frame, blades, and a fusible link. The blades are attached to the frame and are held open by a spring can be opened or closed by a motor or manually, and fusible link actuation for closure control. Its construction is similar to VCD but the difference is that they are supplied with additional Fusible link, and the nylon bushing is replaced with a bronze one that can stand higher temperatures, motorized fire damper actuators offer faster response than MVCD ones, also certified.

Advantages:

- Less restriction of airflow
- Less noise when closing
- More durable

Combination fire and smoke dampers

Combination fire and smoke dampers are designed to close in the event of a fire or smoke event. They are typically more complex than fire dampers only, but they offer a number of advantages, including:

- Ability to isolate both fire and smoke
- Improved fire protection

Combination fire and smoke dampers can be either motorized or manual. Motorized dampers are closed automatically by a motor, while manual dampers must be closed manually. They share the same design as FDB but with additional SST Jamb seal installed at the sides of the damper to avoid smoke leakage, they are also equipped with 2 fusible links (mechanical & Electrical).

Motorized combination fire and smoke dampers

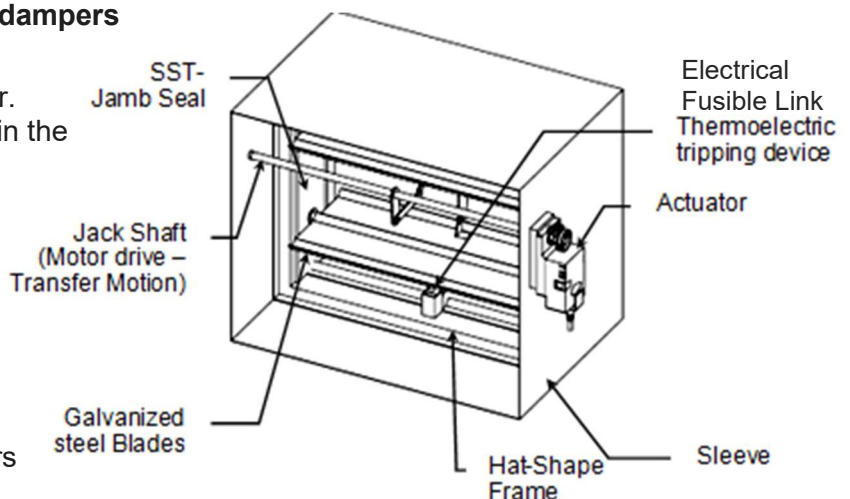
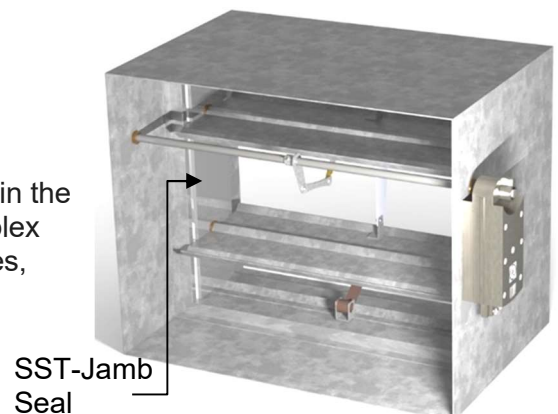
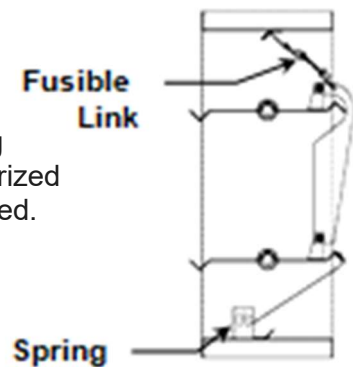
Most common type of combination damper. They are closed automatically by a motor in the event of a fire or smoke event.

Advantages:

- Automatic closure
- Improved fire protection

Disadvantages:

- More expensive than manual dampers
- Can be more difficult to install



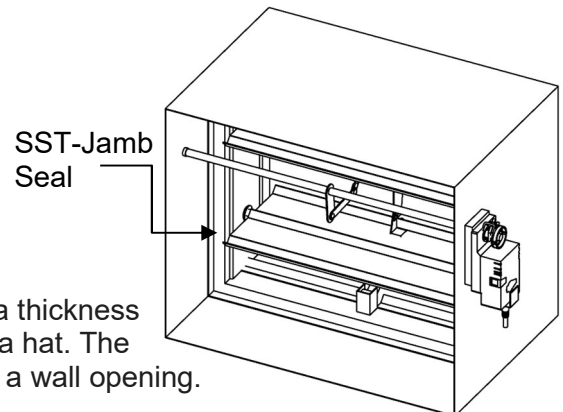
Fire dampers (FDB & FSD) Frames

There are 3 main frame options suitable with various installation requirements:

- Motorized fire damper in sleeve (Type 1)
- Motorized fire damper with flange (Type 2)
- Motorized slip-in damper (Type 3)

Motorized fire damper in sleeve (Type 1)

The frame is a sector that resembles a piece of duct with a thickness of 1.5 mm. Another frame is fixed inside it in the shape of a hat. The blades are placed inside it. This damper is installed inside a wall opening.

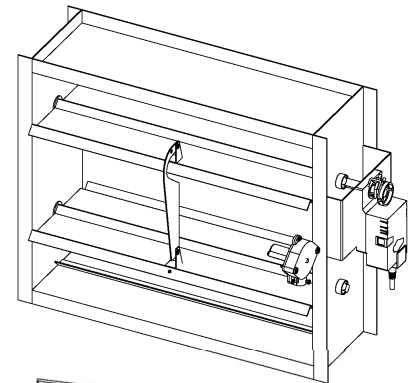


The actuator is not mounted directly on the blade rotation shaft, but is offset from it laterally and connected to it by a link. The electronic melting link is also fixed on the sector.

This type can be ordered with a duct mate flange to connect the duct to it, according to the customer's request.

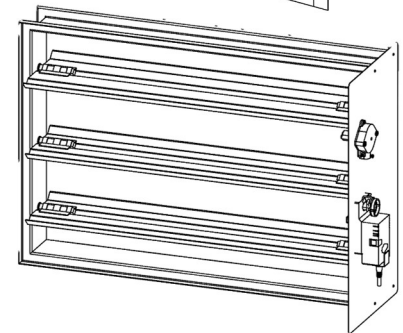
Motorized fire damper with flange (Type 2)

This damper is installed on the duct. It is similar to volume damper with flange, except that bronze bearings must be used instead of nylon. The electronic melting link is supplied separately so that the customer can install it on the duct next to the damper.



Motorized slip-in damper (Type 3)

This damper is a frame in the shape of a hat. A panel is fixed on the side of the frame for the actuator to be mounted on, offset from the main blade rotation shaft. A side opening is made in the duct and the frame is pushed from there into the duct, so that the panel is outside the duct.



Selection and installation

The type of damper that is selected for a particular application will depend on a number of factors, including the size and location & way of installment (Wall, Duct), the required fire rating, and the budget.

Dampers should be installed by a qualified professional to ensure proper installation and operation.

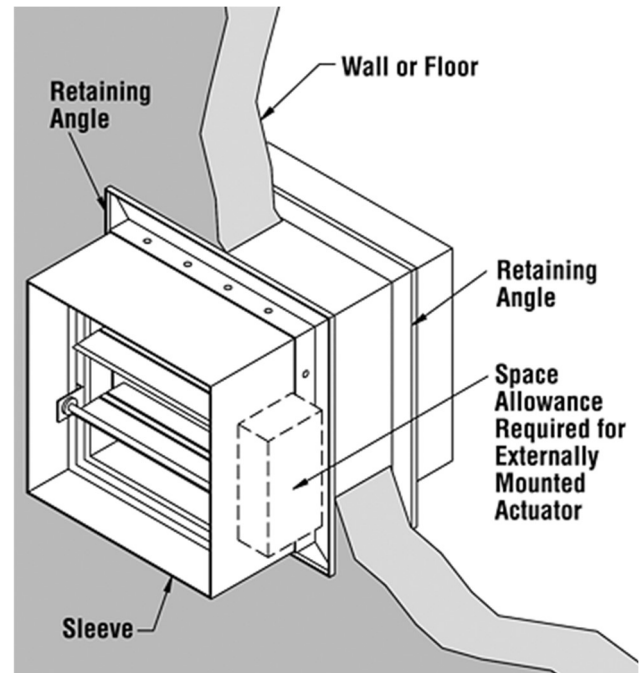
Maintenance

Dampers should be inspected and maintained on a regular basis to ensure that they are in good working order. This includes inspecting the damper for damage, lubricating the moving parts, and replacing the fusible link as needed.

Installation:

The Installation of damper depend on the frame type as mentioned before, usually the fire dampers are supplied with sleeve for wall installation. The sleeve is secured to the wall using retaining angles, riveted to both the damper sleeve and the wall itself. Refer to the figure and QR code for visual clarification.

Flanged FDs installation is similar to VCDF.



How To Order Dampers:

Before requesting a damper quotation, please confirm the following details with the customer:

1. **Damper type:** (GSH, BDD, PRD, VCDH, MVCD, FD, FDB, FSD).
2. **Damper Size:** Width (W) and Height (H), Nominal size must match the duct size.
Minimum sizes are listed in the checklist.
Larger sizes will be supplied as multi-sections.
3. **End Connection type:** (Duct mounting, wall mounting, Flanged, sleeve, duct mate flange, C-Clip, With round or oval connection).
4. **Control:** Manual or Motorized (specify voltage and control type: ON-OFF, 3-P, or Modulating)
5. **Non-Standard Options:**
 - Low-leakage
 - Stainless steel construction
 - Additional sealing
 - Special control features (Actuators, Position indicators,.....)
 - Others requests

Note:

It's important to mention additional cost for any non-standard options.

By gathering this information, you'll ensure accurate damper selection and pricing, streamlining the ordering process.